LIBBY — PATIENT/CAREGIVER TRACK

osteosarcoma-mets-dll3-h7r2

Plain-language summary for patient and caregiver

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What this page is

You (or your clinician) asked Libby to look at one specific situation: **metastatic osteosarcoma where the cancer is showing signs that DLL3 — a protein that some new immunotherapies target — might be present**. Your input was that you'd consider high-risk options if the upside is meaningful.

The DLL3 information you gave us was at the **RNA level**— that tells us the gene is being read, but it doesn't tell us whether the DLL3 protein is actually sitting on the surface of the cancer cells (which is what those new drugs need to grab onto). So the recommendations come in two parts: **a single first step everyone agreed on**, and then **two parallel paths**depending on what that first step finds.

The first step everyone agreed on

Get a DLL3 protein test on your tumor — specifically an IHC stain using an antibody called SP347. Your oncologist can order this on stored tissue from a previous biopsy, or fresh tissue if needed. It's non-toxic, takes 1–3 weeks, and costs almost nothing compared to a treatment cycle. **All five reviewers ranked this as the first action — without dissent.**

The test result then tells you which of the two paths below applies.

Path A — if the DLL3 test comes back POSITIVE (≥1%, ideally ≥25%)

In this branch, a brand-new DLL3-targeted immunotherapy is on the table. This is the path that matches what you said you wanted.

Option A1 — Tarlatamab via clinical trial NCT06788938

The board recommended this as the lead option in this branch.

What it is.A "bispecific T-cell engager" given as an infusion every two weeks. One end of the molecule grips DLL3 on the cancer cell, the other end recruits an immune T cell to attack it. Currently approved for small-cell lung cancer. The trial (run by the University of California Lung Cancer Consortium) accepts patients with **any** DLL3-positive tumor, including osteosarcoma, **as long as the IHC test confirms DLL3 is on the cell surface at sufficient levels**.

What the upside might look like. In small-cell lung cancer, where this drug is approved, tarlatamab nearly halved the risk of death compared to chemotherapy in a recent randomized trial. About 40 of every 100 patients had measurable tumor shrinkage. **Whether this translates to osteosarcoma is exactly the question the trial would answer — and you'd be helping to answer it.**

The main risks.A reaction called "cytokine release syndrome" happens in about half of patients (mostly mild; about 1 in 100 are severe). Neurologic side effects in roughly 10%. The first cycle requires being in hospital for monitoring.

Where the board agreed and where they didn't.

- *Risktaker, advocate, conservative, consensus reviewer:*endorse. The
- *Critic:*still dissents — even with a positive IHC, no osteosarcoma

Option A2 — Regorafenib (backbone if the trial isn't reachable)

What it is.A pill at home, 3 weeks on / 1 week off.

Why it's still here even in the trial-positive branch. Trial slots aren't guaranteed; tolerability isn't guaranteed; some patients prefer to have a non-trial option in their pocket. Two clinical trials in metastatic osteosarcoma showed regorafenib delays cancer growth by about 2 months on average, with manageable side effects (skin reaction on hands and feet, high blood pressure, fatigue).

Option A3 — Cabozantinib

What it is. Another pill at home; targets different growth pathways (MET and VEGFR2).

The trial data.A 42-patient trial showed about 12 of every 100 had tumor shrinkage; about a third had no growth at 6 months.

When to consider it. If regorafenib is contraindicated, has been used, or wasn't tolerated.

Path B — if the DLL3 test comes back NEGATIVE or below threshold

In this branch, the tarlatamab pathway is closed — the drug needs the target on the cell surface, and the IHC told us it isn't there at sufficient density. This is disappointing but actionable.

Option B1 — Regorafenib

The board recommended this as the lead option in this branch.

Same drug as Option A2 above — but here it moves to first place because the trial that outranked it is no longer reachable. All five reviewers endorse. Two replicating clinical trials in your exact situation, modest but real benefit, predictable side-effect profile your oncologist knows how to manage.

Option B2 — Cabozantinib

Same drug as Option A3 above. Same role: alternative if regorafenib is contraindicated, exhausted, or not tolerated.

Option B3 — Search for non-DLL3 trials *(considered with caveats)*

You said you preferred trial-based options. Even with the DLL3 pathway closed, that preference is real. The board surfaced this as a third option in the negative branch: ask your oncologist (or a referral to an academic sarcoma center) about other osteosarcoma-relevant trials currently enrolling — different mechanisms, different targets. Specific trial selection wasn't part of this Libby run, so this is a "next conversation" item rather than a specific recommendation.

What the board could not figure out

Several things weren't in the inputs:

- **Was DLL3 measured at the protein level (IHC) or only RNA?**Resolved by
- **What treatment have you already had?**The recommendations assume
- **What's your performance status (how active are you day-to-day)?**Most
- **Where do you live, and what's your access to academic cancer centers?**

Questions to ask your oncologist

1. **Can we order DLL3 IHC (SP347 antibody) on my tumor?**That's the first
2. **If the IHC comes back positive, what's the path to NCT06788938?**
3. **If the IHC comes back negative, are there other osteosarcoma trials
4. **If we pursue regorafenib (in either branch), what's the plan for
5. **Have I already had regorafenib or cabozantinib? What was the response?**
6. **What's the realistic time-to-response for each option, and how will we
7. **Is there anything in my labs or scans that would change which of these
8. **If the tarlatamab trial is on the table, what's the realistic timeline

Where to read more

- Clinician summary— the technical version of this page.
- Trial table— the trials Libby reviewed.
- Evidence list— the clinical and lab studies cited.
- Tumor-board transcript— what each reviewer actually said.
- Recommendations detail— full structured tables.

Sources

The recommendations cited the following published reports and clinical-trial records:

- Davis et al. *J Clin Oncol*2019 — regorafenib SARC024 RCT in metastatic
- Duffaud et al. *Lancet Oncol*2019 — regorafenib REGOBONE RCT (PMID 30477937)
- Italiano et al. *Lancet Oncol*2020 — cabozantinib CABONE phase-2 in
- Ahn et al. *N Engl J Med*2023 — tarlatamab DeLLphi-301 in SCLC (PMID 37861218)
- Mountzios et al. *N Engl J Med*2025 — tarlatamab DeLLphi-304 in SCLC
- Yao et al. *Oncologist*2022 — DLL3 as a therapeutic target review
- Wang et al. *J Exp Clin Cancer Res*2018 — Notch3/DLL3 axis in osteosarcoma
- Trial registry NCT06788938 — UCCC-01 / UCLA L-10 tarlatamab basket